CHISG: Technical Specification

Contextualised Hierarchical Iterative Semantic Groupings

Author: Michael Stewart

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Contact: michael.stewart@espthinking.co.uk

Live System: https://espthinking.co.uk

What CHISG Is

CHISG is a methodology for structuring knowledge into a machine-readable graph that solves three problems no current system addresses together:

1. **Hallucination Reduction** — Grounding AI responses in verified, sourced knowledge claims
2. **Cross-Domain Connection Discovery** — Automatically detecting structural analogies between disparate fields
3. **Gap Analysis** — Identifying where our collective knowledge has holes, contradictions, or untested assumptions

It is not a product. It is infrastructure — a knowledge-graph methodology that can be applied to any domain where structured, trustworthy knowledge matters.

The Three Core Aims

1. Reducing AI Hallucination

The Problem: Large language models generate plausible but incorrect information. In education, law, medicine, and policy, this is dangerous. Current mitigation strategies (prompt engineering, RLHF, retrieval-augmented generation) reduce hallucination frequency but cannot eliminate it because the underlying knowledge remains unstructured text.

The CHISG Approach: Build a deterministic grounding layer.

Every knowledge claim in CHISG is stored as a **semantic unit**:

[Entity A] —[relation]→ [Entity B]

+ backward relation

+ context[] (source, domain, level, topic, exam board, page)

+ trust score (computed, not assigned)

When an AI system generates a response, CHISG provides:

- **Verification**: Does this claim exist in the graph? With what trust score?
- **Contradiction Detection**: Does the graph contain claims that contradict this?
- **Scope Boundaries**: Is this claim supported at the learner's level, or is it a simplification that breaks at higher levels?
- **Provenance**: Which sources support this claim? Are they independent?

The key insight is that **truth is computed from structure, not declared by authority**. A claim supported by multiple independent sources via multiple logical routes has high structural integrity. A claim supported by one source repeated many times has high popularity but low structural integrity. CHISG distinguishes between these.

Current Implementation: 579 skills, 1,120 semantic links, 27 course definitions stored in Weaviate (vector database). The Go backend generates embeddings and queries via `nearVector` for semantic matching. GCSE Science content (AQA specification) is the initial domain.

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2. Finding Connections Between Disparate Elements

The Problem: Knowledge is taught and stored in silos. Physics concepts are separated from biology concepts, which are separated from psychology concepts. But the underlying structural patterns are often identical. A teacher who sees the connection can use it as an analogy. An AI system without structured knowledge cannot.

The CHISG Approach: Structural analogy detection via relational pattern matching.

Every semantic link has the same structure: `[A] —[relation]→ [B]`. When the same relational pattern appears across different domains, CHISG detects it automatically.

Worked Example:

Domain	Entity A	Relation	Entity B
Physics	Bimetallic strip	differential expansion causes	curvature
Biology	Auxin gradient	differential elongation causes	phototropic bending
Biology	Guard cells	differential turgor causes	stomatal curvature

All three share the identical relational pattern: **differential X causes curvature**. A student who deeply understands the bimetallic strip has already learned the logic pattern for auxin response — they need the domain vocabulary mapped onto the same structural relationship.

Why This Matters:

- **For teaching**: Analogies are the most powerful pedagogical tool. Currently, every analogy must be manually authored by an expert teacher. CHISG makes analogy detection automatic and exhaustive.
- **For research**: Cross-domain structural patterns reveal deep principles. When the same pattern appears in physics, biology, and economics, it points to a fundamental mechanism worth investigating.
- **For AI systems**: Structured analogy retrieval replaces vague "you might also be interested in" recommendations with precise structural matches: "This is like the bimetallic strip you learned in physics — same pattern, different molecules."

Implementation: The `context[]` array on each semantic link includes domain tags. Pattern matching queries identify identical relational structures across different domains:

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3. Gap Analysis: Finding the Cutting Edge of Knowledge

The Problem: We don't know what we don't know. Curriculum content is presented as complete, but every field has gaps — relationships that should exist based on structural patterns but have never been tested, or areas where claims contradict each other without resolution.

The CHISG Approach: Compute what's missing from the graph.

Three types of gap analysis:

3a. Structural Gaps (Missing Links)
If the graph contains `A → B` and `B → C` but not `A → C`, and the relation types are transitive, this is a structural gap. Either:

- The relationship exists but hasn't been captured yet (data gap)
- The relationship doesn't exist, which is itself interesting (research gap)

3b. Analogy Gaps (Incomplete Patterns)

If a relational pattern appears in three domains but is absent from a fourth where it would be expected, this is an analogy gap. Example: "Differential X causes curvature" appears in physics, botany, and cell biology — does it appear in developmental biology? If not, is that because it doesn't apply, or because no one has looked?

3c. Trust Gaps (Low-Confidence Claims)

The trust scoring system identifies claims that are:

- **High popularity, low structural support** — repeated everywhere but with only one logical path (possible echo chamber)
- **High structural support, low popularity** — independently derivable through multiple routes but rarely stated (possible overlooked insight)

- **Contradicted** — one source claims `A causes B`, another claims `A inhibits B` (unresolved contradiction requiring investigation)

The Trust Score Formula:

The Equation

$$\boldsymbol{\tau}\left(\boldsymbol{A} \overset{r}{\rightarrow} \boldsymbol{B}\right) = \boldsymbol{\varphi}(\boldsymbol{M_a}) \;.\left[\mathbf{1} - \mathbf{exp}\left(-\left(\boldsymbol{\alpha}\,\mathbf{ln}(\mathbf{1} + \boldsymbol{N_s}) + \boldsymbol{\beta}\boldsymbol{N_p}\right)\right)\right]$$

The Breakdown (How to explain it)

- $\boldsymbol{\tau}$ **(Trust Score):** A value between 0 and 1 (never exactly 1, adhering to "Honest Uncertainty").
- $\boldsymbol{\varphi}(\boldsymbol{M_a})$ **(The Logic Gate):** The semantic/structural fit. If the analogy is nonsense (0), the entire equation collapses to 0. Logic precedes Data.
- $\mathbf{ln}(\mathbf{1} + \boldsymbol{N_s})$ **(The Data Log):** Represents **Source Count**. We use a logarithm because accumulating citations yields diminishing returns.
- $\boldsymbol{N_p}$ **(The Independence Scalar):** Represents **Independent Path Count**. This is weighted heavily ($\boldsymbol{\beta}$) because independent derivation (e.g., proving a concept via both History and Physics) is the highest form of verification in CHISG.
- $\mathbf{1} - \boldsymbol{e}^{-(\dots)}$ **(The Asymptote):** An exponential saturation function. It ensures that as evidence mounts, we approach "Truth" asymptotically, but we never claim to possess "The Absolute."

The Echo Chamber Filter: If ten sources all cite one original paper, CHISG clusters them as a single opinion node rather than counting them as ten independent confirmations. Source independence is tracked through provenance metadata.